PS6: Identifying structure in derivative problems - Answer key qr

1.

$$f(t) = \frac{7}{t} - 7^t + 3t^7 - 2\sin(t) + \sqrt{t}$$

$$f'(t) = -\frac{7}{t^2} - 7^t \cdot \ln(7) + 21t^6 - 2\cos(t) + \frac{1}{2\sqrt{t}}$$

2.

$$g(x) = (2x^2 - 5x - 3)\cos(x)$$

$$g'(x) = (2x^2 - 5x - 3) \cdot (-\sin(x)) + (4x - 5) \cdot (\cos(x))$$

3.

$$h(y) = \frac{e^y}{3y^2 + 3y - 5}$$

$$h'(y) = \frac{(3y^2 + 3y - 5) \cdot (e^y) - (e^y) \cdot (6y + 3)}{(3y^2 + 3y - 5)^2}$$